Local Spectral Analysis of Saddle-Node and Hopf Bifurcations in the Plant–Kim Neuron Model

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1. Introduction

In 1976, Plant and Kim [1] developed a seven-dimensional dynamical system describing the bursting membrane voltage of the *Aplysia* abdominal ganglion *R15* neuron. The model can also represent the effects of tetrodotoxin (TTX) and ouabain observed experimentally [2]. TTX is modelled by reducing the maximum conductance g_I of the I channel, while ouabain is modelled by reducing the electrogenic pump current I_{ep}. Experimentally, bursting is replaced by a slow wave under TTX, regular beating under ouabain, and a stable equilibrium when both drugs are applied.

We parameterise these effects using $\alpha_I, \alpha_P \in [0, 1]$, with

$$g_I = \alpha_I g_I^{(0)}, \qquad I_{\text{ep}} = \alpha_P I_{\text{ep}}^{(0)}.$$

Previous continuation analysis of these parameters identified biologically relevant saddle-node and Hopf bifurcations, which form the focus of the present local analysis. Subsequent mathematical studies of Plant-type *R15* models have used bifurcation and fast–slow methods to investigate the dynamical mechanisms underlying neuronal bursting [3, 4].

The aim of this study is to independently characterise these bifurcations through analytical Jacobian construction and local spectral analysis. In particular, we test the saddle-node non-degeneracy and local fold geometry, examine the critical Hopf eigenvalues and their oscillation-period prediction, and verify the associated spectral conditions independently of the numerical continuation classifications.

2. Methods

The state vector of the Plant–Kim model is

$$\mathbf{x} = \begin{bmatrix} V & X_K & X_A & Y_A & X_P & X_I & Y_I \end{bmatrix}^\top \in \mathbb{R}^7,$$

with the complete dynamical system provided in Appendix A. We denote the system by $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}; \boldsymbol{\alpha}, \boldsymbol{\mu}_0)$, where $\boldsymbol{\alpha}$ contains the active bifurcation parameters and $\boldsymbol{\mu}_0$ contains the remaining fixed parameters.

2.1. System Jacobian

The full 7×7 Jacobian $\mathbf{J} = D_{\mathbf{x}}\mathbf{F}$ is provided in Appendix B. Each gating equation depends only on the membrane voltage V and its corresponding gating variable, so the lower-right 6×6 gating submatrix is diagonal. Coupling between distinct gates therefore occurs indirectly through V,

while the voltage equation depends on every gating variable through the ionic currents, producing the non-zero entries across the first row of $\mathbf{J}$.

The Jacobian was implemented symbolically in Python using `SymPy` [5]. At each bifurcation landmark, the analytical Jacobian was evaluated numerically and validated against a centred finite-difference approximation, while its eigenvalues were cross-checked against the corresponding `PyDSTool` continuation output. The resulting numerical validation is reported in Appendix C.

2.2. Saddle-node analysis

A saddle-node bifurcation is locally characterised by a simple zero eigenvalue of $\mathbf{J}$, with no other eigenvalues on the imaginary axis. At the saddle-node bifurcation point LP_{bio}, we compute right and left null vectors $\mathbf{q}$ and $\mathbf{p}$ satisfying

$$\mathbf{J}\mathbf{q} = \mathbf{0}, \qquad \mathbf{p}^\top \mathbf{J} = \mathbf{0}, \qquad \mathbf{p}^\top \mathbf{q} = 1.$$

The saddle-node non-degeneracy coefficients are then

$$a = \mathbf{p}^\top \mathbf{F}_{\alpha_I}, \qquad b = \frac{1}{2}\mathbf{p}^\top D^2_{\mathbf{x}}\mathbf{F}[\mathbf{q}, \mathbf{q}],$$

where $a \neq 0$ and $b \neq 0$ are required for a generic fold [6].

A centre-manifold reduction gives the leading-order normal form

$$\dot{u} = a(\alpha_I - \alpha_I^{\text{LP}}) + bu^2 + \cdots,$$

from which the voltage component of the nearby equilibrium branches satisfies

$$V - V^{\text{LP}} \simeq \pm q_V \sqrt{-\frac{a}{b}(\alpha_I - \alpha_I^{\text{LP}})}. \tag{1}$$

Here q_V denotes the voltage component of the critical null vector. The derivation of (1) from the centre-manifold normal form is given in Appendix E. This prediction is compared directly with the numerical equilibrium continuation branch near LP_{bio}.

2.3. Hopf analysis

A Hopf bifurcation is locally associated with an isolated complex-conjugate pair

$$\lambda_\pm(\alpha_P) = r(\alpha_P) \pm i\omega(\alpha_P)$$

crossing the imaginary axis. At the critical value $\alpha_P = \alpha_P^{\text{H}}$,

$$r(\alpha_P^{\text{H}}) = 0, \qquad \omega_{\text{H}} := \omega(\alpha_P^{\text{H}}) \neq 0,$$

while the remaining eigenvalues must remain away from the imaginary axis.

At the Hopf point, the imaginary component of the critical pair determines the linearised

small-amplitude oscillation period,

$$T_{\mathrm{lin}} = \frac{2\pi}{\omega_{\mathrm{H}}},$$

which is compared with periods obtained from the `AUTO` limit-cycle branch [7].

Finally, we test whether the critical pair crosses the imaginary axis transversely through

$$\frac{d}{d\alpha_P} \operatorname{Re} \lambda_{\mathrm{crit}}(\alpha_P)\Big|_{\alpha_P^{\mathrm{H}}} \neq 0.$$

The transversality derivative was estimated using centred finite differences at $\alpha_P^{\mathrm{H}} \pm h$ across several step sizes, with the step-size validation reported in Appendix D.

3. Results

3.1. Saddle-Node Bifurcation under TTX

The saddle-node bifurcation LP_{bio} occurs along the $\alpha_P = 1$ slice at

$$(\alpha_I^{\mathrm{LP}}, V^{\mathrm{LP}}) \approx (0.614, -26.13).$$

The Jacobian spectrum evaluated at the fold is reported in Table 1.

Table 1: Real parts of the Jacobian eigenvalues at LP_{bio}. The highlighted eigenvalue is the critical mode associated with the fold.

Index i	$\mathrm{Re}(\lambda_i)$
1	-2.86722×10^{-1}
2	-9.99896×10^{-2}
3	5.88762×10^{-2}
4	-1.53906×10^{-2}
5	-4.13244×10^{-3}
6	$\mathbf{-1.26588 \times 10^{-10}}$
7	-1.25041×10^{-4}

The critical eigenvalue is $\lambda_{\mathrm{crit}} \approx -1.266 \times 10^{-10}$, while the equilibrium residual is $\|\mathbf{F}(\mathbf{x}^{\mathrm{LP}})\|_\infty = 2.498 \times 10^{-16}$. The closest non-critical eigenvalue to zero has magnitude 1.25×10^{-4}, while one non-critical mode has positive real part $\mathrm{Re}(\lambda_3) \approx 0.0589$.

Figure 1 shows the critical real eigenvalue approaching zero along the local continuation window together with the associated fold in the equilibrium voltage branch.

At the fold, the saddle-node non-degeneracy coefficients were

$$a \approx -0.206, \qquad b \approx 1.15 \times 10^{-3},$$

and the smallest singular value of the Jacobian was $\sigma_{\min}(\mathbf{J}) = 1.20 \times 10^{-13}$.

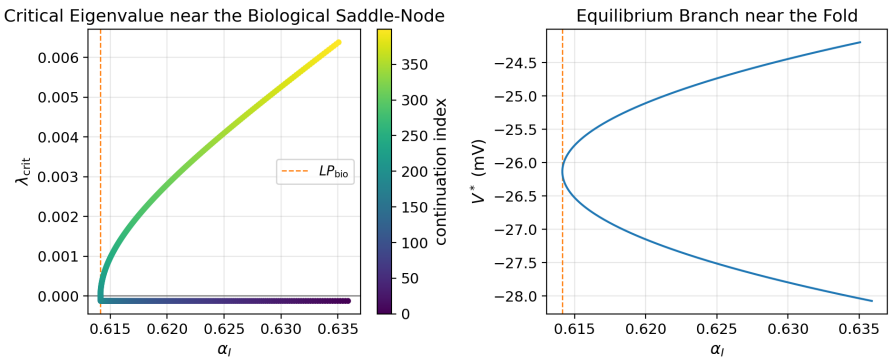


Figure 1: Local spectral behaviour near LP_{bio}. Left: critical real eigenvalue λ_{crit} along the α_I continuation branch, approaching zero at the fold. Right: corresponding fold in the equilibrium voltage branch $V^\star$.

Figure 2 compares the numerical continuation branch with the leading-order centre-manifold prediction

$$V - V^{\mathrm{LP}} \simeq \pm q_V \sqrt{-\frac{a}{b}(\alpha_I - \alpha_I^{\mathrm{LP}})}.$$

The two curves remain closely aligned over the sampled local parameter window.

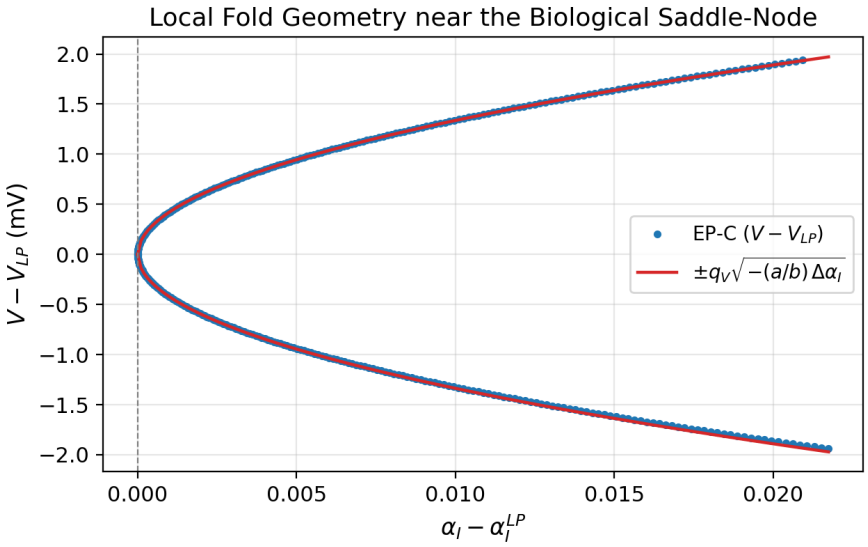


Figure 2: Local fold geometry near LP_{bio}. Numerical equilibrium-continuation data are compared with the leading-order square-root prediction obtained from the centre-manifold reduction.

3.2. Hopf Bifurcation under Complete TTX and Ouabain

Along the complete-TTX slice $\alpha_I = 0$, the Hopf bifurcation H_{bio} occurs at

$$(\alpha_P^{\mathrm{H}}, V^{\mathrm{H}}) \approx (0.2679, -44.79).$$

At this point, the Jacobian possesses the complex-conjugate pair

$$\lambda_{\pm} \approx -6.29 \times 10^{-13} \pm 7.54758 \times 10^{-4} i.$$

The remaining five eigenvalues have negative real parts, with the closest non-critical mode having real part approximately -8.46×10^{-3}.

Across the local continuation window, the real part of the critical pair changes sign through H_{bio}, while its imaginary part remains nonzero, as shown in Figure 3.

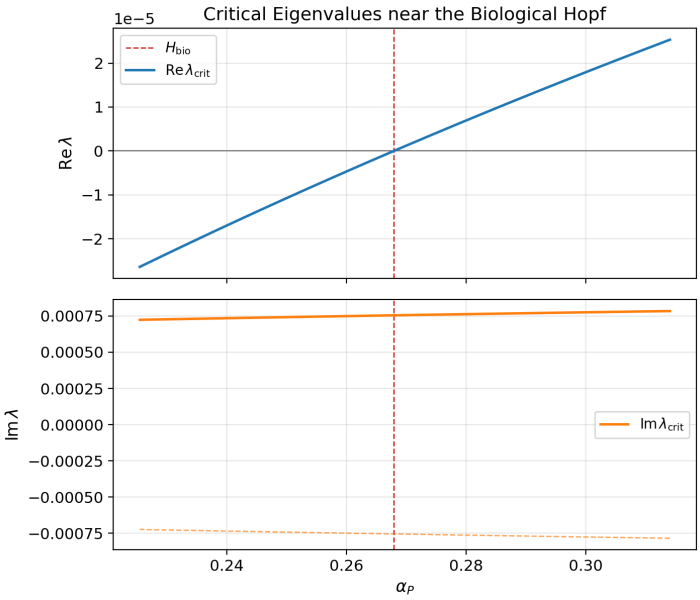


Figure 3: Critical eigenvalue pair near H_{bio}. Top: $\text{Re}(\lambda_{\text{crit}})$ crosses zero at the Hopf threshold (vertical dashed line). Bottom: $\text{Im}(\lambda_{\text{crit}})$ remains nonzero through the crossing.

Using $\omega_{\text{H}} = 7.54758 \times 10^{-4}\,\text{ms}^{-1}$ gives the linearised period prediction

$$T_{\text{lin}} = \frac{2\pi}{\omega_{\text{H}}} = 8324.768\,\text{ms}.$$

For the smallest sampled `AUTO` limit cycle, with voltage amplitude $V_{\max} - V_{\min} \approx 1.41 \times 10^{-3}\,\text{mV}$, the relative discrepancy between $T_{\texttt{AUTO}}$ and T_{lin} was approximately 3.42×10^{-10}. Figure 4 shows the `AUTO` limit-cycle period approaching T_{lin} as the cycle amplitude decreases.

The transversality derivative estimated using centred finite differences remained stable across step sizes $10^{-5} \le h \le 3 \times 10^{-3}$, with

$$\frac{d}{d\alpha_P} \text{Re}\,\lambda_{\text{crit}} \approx 5.83566 \times 10^{-4}.$$

The full step-size calculations are reported in Appendix D.

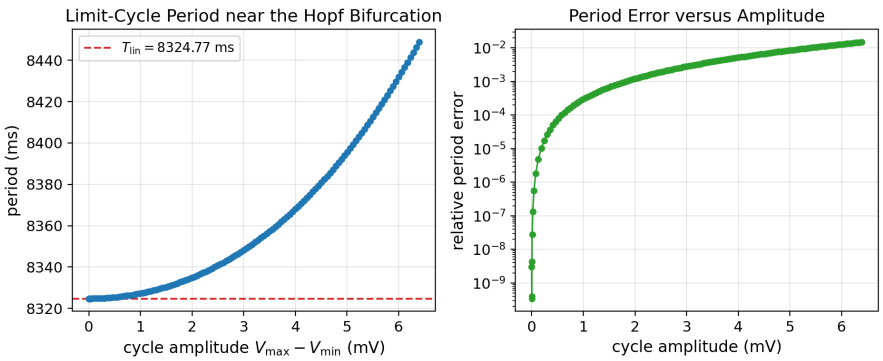


Figure 4: Period convergence near H_{bio}. Left: `AUTO` limit-cycle period against voltage amplitude, with $T_{\text{lin}} = 8324.77\,\text{ms}$ shown as a reference. Right: relative period error on a logarithmic scale as the periodic orbit approaches the Hopf point.

4. Discussion

The local spectral and normal-form analyses provide complementary mathematical explanations for the two bifurcations identified by numerical continuation. At LP_{bio}, the saddle-node is characterised by a simple real eigenvalue approaching zero, together with nonzero parameter and quadratic non-degeneracy coefficients. In contrast, H_{bio} is generated by an isolated complex-conjugate pair crossing the imaginary axis transversely. These results therefore distinguish the local mechanisms underlying the two continuation labels rather than relying solely on their numerical detection.

At LP_{bio}, the vanishing critical eigenvalue, near-singular Jacobian, and nonzero coefficients a and b are all consistent with a generic saddle-node bifurcation. However, the full spectrum in Table 1 also shows that a non-critical eigenvalue already has positive real part, with $\text{Re}(\lambda_3) \approx 0.0589$. Hence the equilibrium is already unstable independently of the fold direction, and the saddle-node occurs between unstable equilibrium branches. This is important for interpreting the biological TTX transition, as although LP_{bio} identifies a genuine local change in equilibrium geometry, it does not by itself provide a complete explanation for the slow-wave behaviour observed under TTX. The close agreement between the numerical continuation branch and the leading-order square-root approximation in Figure 2 nevertheless supports the centre-manifold reduction as an accurate local description of the fold geometry near LP_{bio}.

The Hopf bifurcation provides a different local mechanism. At H_{bio}, the critical complex-conjugate pair crosses the imaginary axis while the remaining eigenvalues remain in the left half-plane, producing a local change in equilibrium stability through an oscillatory mode. The imaginary part of this pair predicts the small-amplitude oscillation period through $T_{\text{lin}} = 2\pi/\omega_{\text{H}}$, and the agreement between this prediction and the `AUTO` limit-cycle period becomes increasingly close as the cycle amplitude tends to zero. This convergence provides an independent link between the local Jacobian spectrum and the periodic-orbit branch detected by continuation. The nonzero derivative $\frac{d}{d\alpha_P}\,\text{Re}\,\lambda_{\text{crit}}$ further supports a transverse Hopf crossing.

A strength of the present analysis is that the bifurcation classifications are verified independently using several complementary calculations. The saddle-node is supported by its spectral signature, Jacobian singularity, non-degeneracy coefficients, and agreement with the predicted square-root fold geometry. Likewise, the Hopf point is supported by the critical eigenvalue pair, the period prediction, convergence of the nearby `AUTO` limit cycles, and the transversality calculation. These checks reduce reliance on continuation labels alone. More broadly, the methods used here and numerical continuation address different aspects of the dynamics: continuation describes the organisation of equilibrium and periodic-orbit branches over parameter ranges, whereas Jacobian, eigenvalue, and normal-form calculations identify the local mathematical mechanisms responsible for changes in stability and bifurcation.

The main limitation of the analysis is its inherently local nature. Linearisation describes dynamics only in a neighbourhood of an equilibrium, while the saddle-node normal form is only expected to approximate the equilibrium branch sufficiently close to LP_{bio}. This locality is visible directly in the results. The Hopf period prediction agrees most closely with the `AUTO` limit cycles as their amplitude approaches zero, while the discrepancy increases for larger cycles. Similarly, the square-root approximation reproduces the local geometry of the saddle-node but is not intended as a global description of the equilibrium branch. Consequently, the present calculations do not by themselves explain the full large-amplitude bursting and slow-wave dynamics of the Plant–Kim model, and should therefore be interpreted alongside continuation and time-domain simulations.

There are also several mathematical extensions that could strengthen the analysis. For the Hopf bifurcation, computing the first Lyapunov coefficient would determine whether the bifurcation is supercritical or subcritical and would provide further information about the stability of the emerging periodic orbit. A fast–slow decomposition, following the geometric approach used for Plant-type bursting models by Rinzel and Lee [4], could help determine how the local bifurcations identified here connect to the full large-amplitude bursting dynamics. Further two-parameter continuation of saddle-node and Hopf loci in (α_I, α_P) could also be used to search for organising codimension-two structures or to determine how robust the observed transitions are to simultaneous variation in both drug parameters. More generally, continuation in additional conductance or time-scale parameters could test whether the local mechanisms found here persist under perturbations of the original Plant–Kim parameter set.

Previous mathematical studies of Plant-type neuronal models have emphasised bifurcation and fast–slow approaches to explaining bursting dynamics. Plant [3] used bifurcation analysis to investigate transitions between oscillatory states, while Rinzel and Lee [4] decomposed the Plant model into fast and slow subsystems to explain how their interaction generates bursting. Later *R15* models incorporated additional ionic and intracellular calcium mechanisms [8]. The present analysis is complementary to these approaches: rather than seeking a global explanation of the entire burst cycle, it characterises the local spectral and normal-form mechanisms at drug-induced saddle-node and Hopf landmarks in the full system. A natural extension would therefore be to combine the present local analysis with a fast–slow decomposition to determine how these bifurcations participate in the global bursting and slow-wave dynamics.

5. Conclusion

This study achieved its aim of independently characterising the local mathematical mechanisms underlying the biologically relevant saddle-node and Hopf bifurcations in the Plant–Kim neuron model. Through analytical Jacobian construction, spectral analysis, normal-form calculations, and comparison with numerical continuation, the saddle-node at LP_{bio} was shown to exhibit the expected generic fold structure, while the Hopf bifurcation at H_{bio} was identified as a transverse oscillatory instability whose critical eigenvalues accurately predict the small-amplitude limit-cycle period. Together, these results provide independent mathematical support for the continuation classifications and demonstrate how local spectral and normal-form analysis can complement numerical continuation by explaining the mechanisms responsible for changes in equilibrium stability and oscillatory behaviour in the model.

A. Plant–Kim R15 Model Equations

The full seven-dimensional Plant–Kim R15 model underlying the analysis in this report is given below. With state vector

$$\mathbf{x} = \begin{bmatrix} V & X_K & X_A & Y_A & X_P & X_I & Y_I \end{bmatrix}^\top,$$

the system is written as $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}; \boldsymbol{\alpha}, \boldsymbol{\mu}_0)$. The untreated model corresponds to $\alpha_I = \alpha_P = 1$, while TTX and ouabain are represented through $g_I = \alpha_I g_I^{(0)}$ and $I_{\text{ep}} = \alpha_P I_{\text{ep}}^{(0)}$, respectively.

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}; \boldsymbol{\alpha}, \boldsymbol{\mu}_0) = \begin{bmatrix} F_1(\mathbf{x}) \\ F_2(\mathbf{x}) \\ F_3(\mathbf{x}) \\ F_4(\mathbf{x}) \\ F_5(\mathbf{x}) \\ F_6(\mathbf{x}) \\ F_7(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \frac{1}{C}\Big[(g_I X_I^3 Y_I + g_T)(V_I - V) + (g_K X_K^4 + g_A X_A Y_A + g_P X_P)(V_K - V) + g_L(V_L - V) + I_{\text{ep}} + I_{\text{ext}}\Big] \\ \frac{1}{\tau_{X_K}(V)}\big[S_K(V) - X_K\big] \\ \frac{1}{\tau_{X_A}}\big[S_A(V) - X_A\big] \\ \frac{1}{\tau_{Y_A}}\big[Z_A(V) - Y_A\big] \\ \frac{1}{\tau_{X_P}}\big[S_P(V) - X_P\big] \\ \frac{1}{\tau_{X_I}(V)}\big[S_I(V) - X_I\big] \\ \frac{1}{\tau_{Y_I}(V)}\big[Z_I(V) - Y_I\big] \end{bmatrix}$$

B. Full 7×7 Jacobian

For the generic gating equation

$$\dot{X} = \frac{X_\infty(V) - X}{\tau_X(V)},$$

it can be readily seen that the derivative with respect to its own gating variable is

$$\frac{\partial \dot{X}}{\partial X} = -\frac{1}{\tau_X(V)}.$$

These correspond to the negative diagonal terms from rows 2 to 7. If we differentiate the same equation with respect to the membrane voltage V, we obtain

$$\frac{\partial \dot{X}}{\partial V} = \frac{X'_\infty(V)}{\tau_X(V)} - \frac{\big[X_\infty(V) - X\big]\tau'_X(V)}{\tau_X^2(V)},$$

and for gates whose time constants are constant, the equation is simplified as

$$\frac{\partial \dot{X}}{\partial V} = \frac{X'_\infty(V)}{\tau_X(V)}.$$

We also obtain the same form at an equilibrium $\mathbf{x}^\star$, as by definition, when $\dot{X} = 0$ we have $X = X_\infty(V)$ and thus the second term vanishes even if τ_X is voltage-dependent.

The Jacobian $\mathbf{J}$ used throughout this report is given in complete detail as

$$\begin{bmatrix} \frac{-X_A Y_A g_A - X_I^3 Y_I g_I - X_K^4 g_K - X_P g_P - g_L - g_T}{C} & \frac{4X_K^3 g_K (V_K - V)}{C} & \frac{Y_A g_A (V_K - V)}{C} & \frac{X_A g_A (V_K - V)}{C} & \frac{g_P (V_K - V)}{C} & \frac{3X_I^2 Y_I g_I (V_I - V)}{C} & \frac{X_I^3 g_I (V_I - V)}{C} \\ -\frac{(S_K(V) - X_K)\frac{d}{dV}\tau_{X_K}(V)}{\tau_{X_K}^2(V)} + \frac{\frac{d}{dV}S_K(V)}{\tau_{X_K}(V)} & -\frac{1}{\tau_{X_K}(V)} & 0 & 0 & 0 & 0 & 0 \\ \frac{\frac{d}{dV}S_A(V)}{\tau_{X_A}} & 0 & -\frac{1}{\tau_{X_A}} & 0 & 0 & 0 & 0 \\ \frac{\frac{d}{dV}Z_A(V)}{\tau_{Y_A}} & 0 & 0 & -\frac{1}{\tau_{Y_A}} & 0 & 0 & 0 \\ \frac{\frac{d}{dV}S_P(V)}{\tau_{X_P}} & 0 & 0 & 0 & -\frac{1}{\tau_{X_P}} & 0 & 0 \\ -\frac{(S_I(V) - X_I)\frac{d}{dV}\tau_{X_I}(V)}{\tau_{X_I}^2(V)} + \frac{\frac{d}{dV}S_I(V)}{\tau_{X_I}(V)} & 0 & 0 & 0 & 0 & -\frac{1}{\tau_{X_I}(V)} & 0 \\ -\frac{(Z_I(V) - Y_I)\frac{d}{dV}\tau_{Y_I}(V)}{\tau_{Y_I}^2(V)} + \frac{\frac{d}{dV}Z_I(V)}{\tau_{Y_I}(V)} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{\tau_{Y_I}(V)} \end{bmatrix}.$$

C. Jacobian Validation Results

The validation is summarised in Table 2, including the maximum absolute residual at the equilibrium point $\|\mathbf{F}\|_\infty$, the maximum absolute difference between the symbolic and finite difference Jacobian matrices $\max \|\mathbf{J}_{\text{sym}} - \mathbf{J}_{\text{FD}}\|$, the relative error using the Frobenius norm $\frac{\|\mathbf{J}_{\text{sym}} - \mathbf{J}_{\text{FD}}\|_F}{\|\mathbf{J}_{\text{FD}}\|_F}$, and the maximum discrepancy between the eigenvalues computed symbolically against those computed by `PyDSTool` [9]. The validation results show close agreement between the symbolic, finite-difference, and continuation calculations, with discrepancies of order 10^{-10} or smaller for the reported comparisons.

Table 2: Numerical validation of Jacobian $\mathbf{J}$ against centred finite-difference approximations and `PyDSTool`.

Point	$\|\mathbf{F}\|_\infty$	$\max \|\mathbf{J}_{\text{sym}} - \mathbf{J}_{\text{FD}}\|$	$\frac{\|\mathbf{J}_{\text{sym}} - \mathbf{J}_{\text{FD}}\|_F}{\|\mathbf{J}_{\text{FD}}\|_F}$	`PyDSTool` eigenvalue mismatch
H_{bio}	3.469×10^{-17}	4.771×10^{-11}	5.704×10^{-11}	1.685×10^{-10}
LP_{bio}	2.498×10^{-16}	6.390×10^{-11}	1.075×10^{-11}	3.176×10^{-10}

D. Transversality Validation

To assess the numerical robustness of the Hopf transversality calculation, the derivative of the critical real eigenvalue was estimated using the centred finite-difference approximation

$$D_h = \frac{\operatorname{Re}\lambda_{\text{crit}}(\alpha_P^{\text{H}} + h) - \operatorname{Re}\lambda_{\text{crit}}(\alpha_P^{\text{H}} - h)}{2h}.$$

Table 3 reports the resulting estimates across several step sizes, together with the critical eigenvalue evaluated on either side of the Hopf point.

Table 3: Centred finite-difference validation of the Hopf transversality derivative. Estimates of $d\,\mathrm{Re}(\lambda_{\mathrm{crit}})/d\alpha_P$ are shown across step sizes h, together with the critical eigenvalue evaluated at $\alpha_P^{\mathrm{H}} \pm h$.

h	$\frac{d\,\mathrm{Re}(\lambda_{\mathrm{crit}})}{d\alpha_P}$	$\mathrm{Re}\,\lambda_{\mathrm{crit}}(\alpha_P^{\mathrm{H}}+h)$	$\mathrm{Re}\,\lambda_{\mathrm{crit}}(\alpha_P^{\mathrm{H}}-h)$	$\mathrm{Im}\,\lambda_{\mathrm{crit}}(\alpha_P^{\mathrm{H}}+h)$	$\mathrm{Im}\,\lambda_{\mathrm{crit}}(\alpha_P^{\mathrm{H}}-h)$
1.00×10^{-5}	5.84×10^{-4}	5.83×10^{-9}	-5.84×10^{-9}	7.55×10^{-4}	7.55×10^{-4}
3.00×10^{-5}	5.84×10^{-4}	1.75×10^{-8}	-1.75×10^{-8}	7.55×10^{-4}	7.55×10^{-4}
1.00×10^{-4}	5.84×10^{-4}	5.83×10^{-8}	-5.84×10^{-8}	7.55×10^{-4}	7.55×10^{-4}
3.00×10^{-4}	5.84×10^{-4}	1.75×10^{-7}	-1.75×10^{-7}	7.55×10^{-4}	7.55×10^{-4}
1.00×10^{-3}	5.84×10^{-4}	5.83×10^{-7}	-5.84×10^{-7}	7.55×10^{-4}	7.54×10^{-4}
3.00×10^{-3}	5.84×10^{-4}	1.74×10^{-6}	-1.76×10^{-6}	7.57×10^{-4}	7.53×10^{-4}

E. Derivation of the local saddle-node fold approximation

Near LP_{bio}, the centre-manifold normal form is

$$\dot{u} = a\Delta\alpha_I + bu^2 + \mathcal{O}(u^3), \qquad \Delta\alpha_I = \alpha_I - \alpha_I^{\mathrm{LP}}.$$

Setting $\dot{u} = 0$ gives the leading-order equilibrium branches

$$u_\pm \simeq \pm\sqrt{-\frac{a}{b}\Delta\alpha_I}.$$

Since nearby states satisfy

$$\mathbf{x} - \mathbf{x}^{\mathrm{LP}} = u\mathbf{q} + \mathcal{O}(u^2, \Delta\alpha_I),$$

the voltage component obeys

$$V - V^{\mathrm{LP}} \simeq \pm q_V\sqrt{-\frac{a}{b}(\alpha_I - \alpha_I^{\mathrm{LP}})}.$$

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